



ITI · AI AGENTS DEVELOPMENT USING PYTHON TRACK

Generative AI & Prompt Engineering

Eng. Fatma Elraey*Lab 1 — Hands-On Prompt Engineering (Lab Assignment)*

Field	Detail
Instructor	Fatma Elraey
Course	Introduction to Generative AI and Prompt Engineering (1-Day Course)
Session	Lab — 3 Hours (Lab Assignment: 100% of grade)
Focus	Writing and refining real prompts in an AI chat tool of your choice — no coding required.

Before You Start

Open any Generative AI chat tool you have access to (ChatGPT, Gemini, Claude, Copilot, or similar — a free account is enough). For each exercise, type the prompt in, read the response, and jot down a short note on what worked or didn't. There's no single "correct" response — you're practicing how to ask well, not memorizing an answer key.

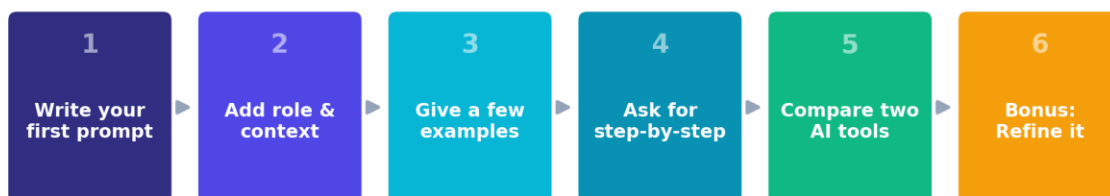


Figure L.1 — Six small steps: a first prompt, then role, examples, reasoning, comparison, and a final refinement.



✓ QUICK CHECK — Before You Start

Q1. What are the four parts of the R-T-C-F prompt framework?

Answer: *Role, Task, Context, Format.*

Q2. What is few-shot prompting?

Answer: *Giving the AI a couple of examples of the format you want before asking for the real thing.*

Q3. What should you do if a response disappoints you?

Answer: *Tell the AI specifically what was wrong and ask it to refine the answer, rather than starting over.*



01 Write Your First Prompt

Start simple. Ask an AI chat tool to explain a topic you already know a little about, then look closely at what makes the response good or not.

- *Pick any topic you know a bit about (a hobby, a subject you studied, a place you've visited).*
- *Type a short, plain prompt asking the AI to explain it in 3-4 sentences.*
- *Read the response and note: did it sound accurate? Was it too generic, or specific enough?*
- *Now ask a follow-up in the same chat: "Make that shorter and friendlier." Notice how it adjusts.*

Why this matters: Almost every AI conversation is iterative — you rarely get the perfect answer on the very first try, and that's completely normal, not a sign you did something wrong.

Try this:

PROMPT

Explain [your topic] in 3-4 sentences,
in plain language.

FOLLOW-UP PROMPT

Make that shorter and friendlier.

02

Add a Role and Some Context

Now compare a plain prompt against the same request with a role and context added, and see how much the response changes.

- *First, ask plainly: "Give me tips for staying productive."*
- *Then try the same request with a role and context added (see the starter prompt below).*
- *Compare the two responses side by side — which one feels more useful to you specifically, and why?*

Why this matters: Adding a role ("You are a...") and context (who it's for, what situation) is usually the single biggest improvement you can make to a vague prompt.

Try this:



PLAIN PROMPT

Give me tips for staying productive.

PROMPT WITH ROLE + CONTEXT

You are a productivity coach. I'm a university student juggling 5 courses and a part-time job. Give me 5 realistic productivity tips for my situation, not generic advice.



03

Give a Few Examples (Few-Shot Prompting)

Sometimes it's easier to show the AI the format you want instead of describing it in words. Practice few-shot prompting by giving 2 examples before asking for more.

- Copy the starter prompt below, which shows 2 example product taglines before asking for 3 more.
- Run it, then try removing the 2 examples and asking for the same 3 taglines with just a plain instruction.
- Compare: did having examples change the style or quality of what came back?

Why this matters: Few-shot prompting is especially useful for matching a specific tone, format, or structure — words alone often aren't as precise as a couple of good examples.

Try this:

PROMPT

Here are 2 taglines for eco-friendly products:

1. "Small changes, big planet."
2. "Kind to earth, kind to you."

Write 3 more taglines in the same short, punchy style, for a reusable water bottle brand.

04

Ask for Step-by-Step Reasoning

For anything with multiple steps or logic involved, asking the AI to reason out loud usually gives a more reliable — and more checkable — answer.

- Give the AI the word problem in the starter prompt below, without asking for steps, and note the answer.
- Now ask the same question again, but add "Think step by step and show your work."
- Compare the two responses — is the step-by-step version easier to verify or trust?

Why this matters: Chain-of-thought prompting doesn't just look more thorough — it measurably reduces careless mistakes on multi-step math and logic problems.

Try this:

PROMPT

A store sells notebooks for \$3 each. If you buy 5 or more, you get a 20% discount on the whole order. How much do 7 notebooks cost?

FOLLOW-UP

Think step by step and show your work.



Answer to check against:

```
7 x $3 = $21  
20% discount: $21 x 0.20 = $4.20  
Final price: $21 - $4.20 = $16.80
```



05

Compare Two AI Tools

Different Generative AI tools can give noticeably different answers to the exact same prompt. Try this out for yourself with a slightly opinion-based question.

- *Pick any two AI tools you can access (e.g. ChatGPT and Gemini, or ChatGPT and Copilot).*
- *Ask both the exact same prompt from below, word for word.*
- *Compare: Did they agree? Was one clearer, more structured, or more helpful than the other?*
- *Write 2-3 sentences summarizing which response you preferred and why.*

Why this matters: No single AI tool is best at everything — building the habit of comparing responses, especially for anything important, is one of the most useful skills from today's lab.

Try this:

PROMPT

What are the 3 most important skills for someone starting a career in AI, and why?



Bonus Stretch Goal — Refine a Weak Prompt

Take a genuinely weak prompt and rebuild it into a strong one using the full R-T-C-F framework from the lecture — Role, Task, Context, Format.

- Start with the weak prompt below and run it as-is. Notice how generic the response feels.
- Now rewrite it using all four R-T-C-F pieces: a role, a specific task, real context, and a clear format.
- Run your improved version and compare it to the original response.
- Optional: pick a task from your own life (studying, a hobby, a project) and write your own R-T-C-F prompt from scratch.

Why this matters: This exact refine-and-compare habit — starting rough, then deliberately adding role, task, context, and format — is the single most transferable skill from today, useful in any AI tool you'll touch in future tracks.

Starter template:

```
WEAK PROMPT
Help me with my resume.

YOUR TURN -- rewrite it using R-T-C-F:
Role:    "You are a..."
Task:    what exactly do you want done?
Context: who is this for? what situation?
Format:  bullet points? a table? a specific length?
```

Nice work! You now have a real, transferable prompt-engineering toolkit — take it with you into every track that follows.